

Assignment 1

Adhoc Wireless Networks (ECEN 4127) [Each question has 4 marks]

MODULE-I [10L]

Introduction [2L]

Ad hoc wireless networks, Applications of Ad hoc wireless networks. Issues in Ad hoc wireless networks, Static and mobile Ad hoc network, Indoor Outdoor network model.

MAC Protocols [8L]

Issues in designing a MAC protocol for Ad hoc wireless Networks, Hidden and Exposed terminal problem, Contention based protocols with reservation mechanisms and scheduling mechanisms, MAC protocols using directional antennas, IEEE802.11 in Ad hoc mode.

1. What is MANET? Compare MANET and WSN.
2. What is VANET? What are the applications of VANET?
3. List any four applications of Adhoc wireless network.
4. State the important characteristics of Adhoc wireless network. Or
Discuss the basic concepts of Adhoc wireless network.
5. Explain the major design issues in Adhoc wireless network.
6. State the challenges due to dynamic topology in Adhoc wireless network.
7. Why is multi hop wireless communication required for Adhoc wireless network?
8. State and explain the differences between infrastructure networks or cellular network and ad hoc wireless networks.
9. What are the differences between ad hoc and sensor wireless networks?
10. What is Data-Centric Network? Explain with diagram.
11. How the deployment scenarios and constraints will change from an indoor model to outdoor model?
12. What are the criteria for designing wireless sensor network?
13. What are the advantages of sensor networks?
14. Design the network architecture for hybrid ad hoc network for mobile nodes.
15. Draw network topology and architecture for static and mobile ad hoc network.
16. What are the difference between static and dynamic as hoc networks?
17. Write down the issues of designing a MAC protocol for Ad-hoc networks.

18. Explain hidden and exposed problem in wireless ad-hoc networks.
19. Describe the common method used in alleviating the hidden terminal problem at the MAC layer.
20. How MACA avoids the hidden and exposed terminal problem? Explain with diagrams.
21. What is contention based protocol? How many types of contention based protocols are there? What is the difference between them?
22. What are the advantages of reservation based MAC protocols over contention based MAC protocols?
23. What are the major differences between contentions based MAC protocol with reservation and scheduling mechanism?
24. How MACA protocol with piggy-backed reservation works? Discuss with an example.
25. Point out the main differences of MACAW with MACA.
26. How does MACAW protocol improve the synchronization with respect to MACA protocol? Explain clearly.
27. How does packet transfer take place using MACA-BI protocol? Explain with a suitable diagram.
28. Explain, how directional antenna based MAC protocol reduces the interference?
29. Outline how node scheduling is done in contention-based MAC protocols with scheduling mechanism.
30. Explain the Directional Busy Tone based MAC Protocol with an example
31. What are the advantages of the BTMA protocol? How are they overcome in the DBTMA protocol?